ADB

Extended Annual Review Report

Project Number: 50248-001
Loan Number: 3495
October 2020

Sunseap Asset (Cambodia) Co. Ltd.
Cambodia Solar Power Project
(Cambodia)

Asian Development Bank

CURRENCY EQUIVALENTS

Currency unit – riel (KR)

		At Appraisal	At Operations Evaluation
		15 July 2016	18 June 2020
$1.00	–	KR4,006.3	KR4,061.0

ABBREVIATIONS

ADB	–	Asian Development Bank
BEP	–	Basic Energy Plan
CFPS	–	Canadian Climate Fund for the Private Sector in Asia
COD	–	commercial operations date
COVID-19	–	coronavirus disease
DSRA	–	debt service reserve account
E&S	–	environmental and safety
EAC	–	Electricity Authority of Cambodia
EDC	–	Electricité du Cambodge
EHSS	–	environment, health, safety, and social
EIRR	–	economic internal rate of return
EPC	–	engineering, procurement, and construction
ESMP	–	environmental and social management plan
FIRR	–	financial internal rate of return
IESE	–	initial environmental and social examination
Lao PDR	–	Lao People’s Democratic Republic
LTA	–	lender's technical advisor
MME	–	Ministry of Mines and Energy
O&M	–	operation and maintenance
PDP	–	Power Development Plan
PPA	–	power purchase agreement
REE	–	rural electricity enterprise
RRP	–	report and recommendation of the President
SEZ	–	special economic zone
WACC	–	weighted average cost of capital

WEIGHTS AND MEASURES

GWh	–	gigawatt-hour
kV	–	kilovolt
kWh	–	kilowatt-hour
MW	–	megawatt
tCO_2	–	ton of carbon dioxide

NOTE

In this report, "$" refers to United States dollars.

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CONTENTS

BASIC DATA
Cambodia Solar Power Project
(LN3495–Cambodia)

Key Project Data	As per ADB Project Documents ($ million)	Actual ($ million)
Total Project Cost	13.20	13.30
ADB Loan		
Committed	3.60	3.25
Disbursed	3.60	3.25
Outstanding		3.00
B Loans (Commercial Banks)		
Committed	3.00	2.70
Disbursed	3.00	2.70
Outstanding		2.41
Official Cofinancing (Canadian Climate Fund for the Private Sector in Asia)		
Committed	3.25	3.25
Disbursed	3.25	3.25
Outstanding		3.00
Debt–Equity Ratio at Completion	2.94x	2.24x

ADB = Asian Development Bank.
Note: Actual figures are based on the exchange rate at appraisal in the report and recommendation of the President. Outstanding figures are based on 29 February 2020 data.

Key Dates	Expected	Actual
Concept Clearance Approval	25 July 2016	25 July 2016
Fact-Finding Mission	September 2016	September 2016
Board Approval	7 December 2016	7 December 2016
Execution of Agreements, Including Risk Sharing	4 April 2017	4 April 2017
First Disbursement	17 May 2017	17 May 2017
Commercial Operations Date	29 August 2017	1 October 2017

Project Administration and Monitoring	Number of Missions	Number of Person-Days
Due Diligence and Appraisal	2	22
Negotiations and Signing	1	7
Monitoring	2	4
XARR Mission	0	0

XARR = extended annual review report.
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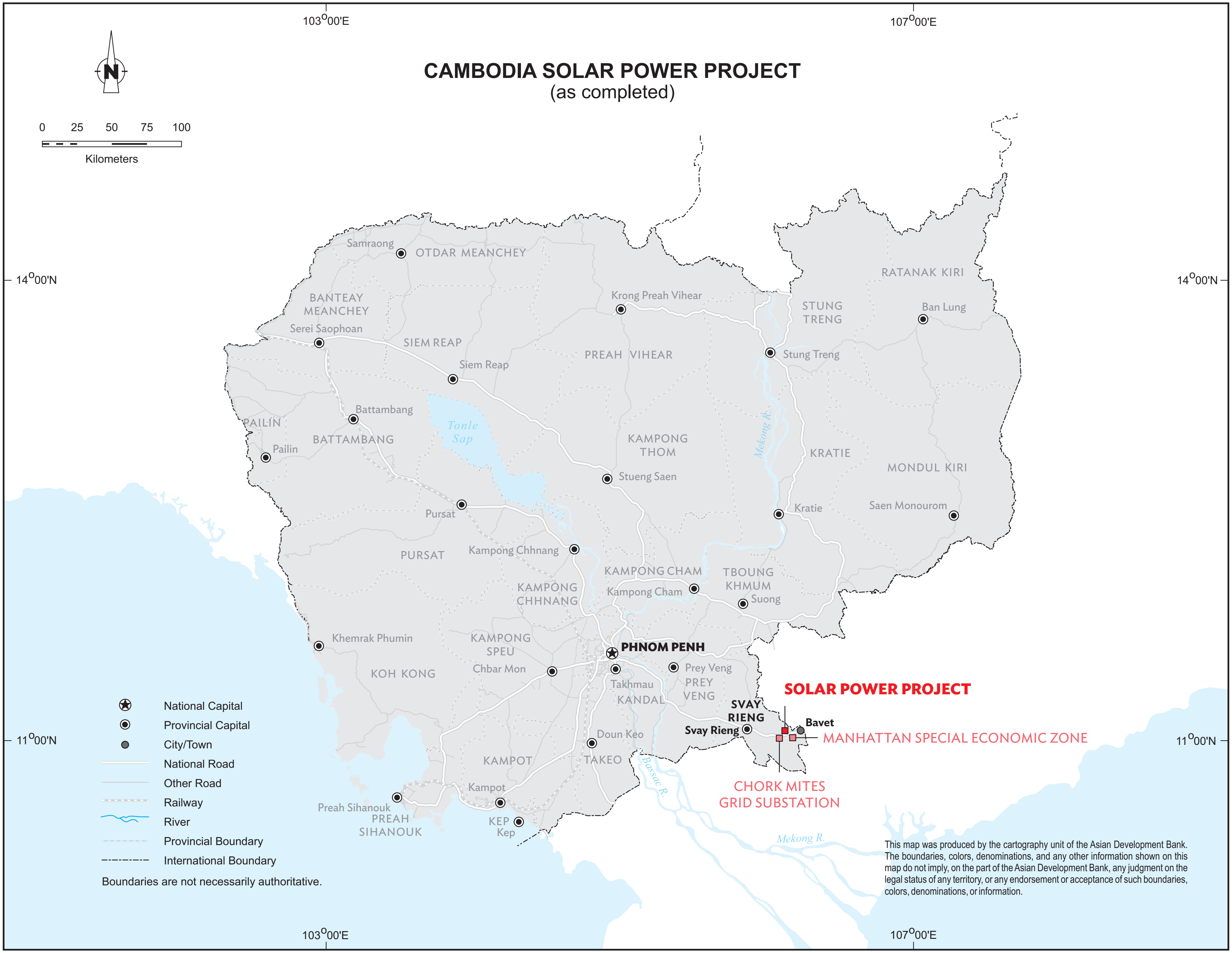

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EXECUTIVE SUMMARY

In December 2016, the Board of Directors of the Asian Development Bank (ADB) approved a debt of $9.85 million to Sunseap Asset (Cambodia) Co. Ltd. to build a 10.5-megawatt solar power plant in Bavet, Cambodia. The approved ADB financing consisted of a loan of $3.60 million from ADB's ordinary capital resources, a B loan of $3.00 million, and concessional financing of $3.25 million.

The project was the first utility-scale solar photovoltaic project in Cambodia. Its implementation proved that a competitively auctioned, privately developed, utility-scale solar plant was both viable and affordable for Cambodia's consumers. The project's implementation clarified the multiple contractual and institutional issues for a private player attempting to develop a utility-scale solar project in Cambodia. This experience paved the way for several solar projects. The project also provided significant economic benefits to Cambodia, including: alleviated the pressing electricity shortage in Bavet, reduced greenhouse gas emissions, electricity imports from Viet Nam, and electricity purchase costs for Electricité du Cambodge.

This evaluation rates the project's overall development outcome as *successful*. The project was completed on time and within budget. Its generation has exceeded appraisal estimates and plant operations have been stable since commissioning.

The review rates the project's contributions to private sector development as *excellent*. Sunseap met most targets on electricity generation, greenhouse gas emission reductions, job creation, and local goods procurement. More importantly, (i) the project's implementation improved the quality and reliability of power supply for consumers in Bavet, which in turn supported the growth of export-oriented manufacturing in the Svay Rieng province; (ii) the project's power purchase agreement and contract structure were replicated in larger solar projects that set record low tariffs in Southeast Asia; (iii) the project demonstrated that long-term project finance was available for Cambodian infrastructure projects.

The project is rated *satisfactory* for economic performance. Economic benefits include (i) the avoidance of diesel and coal-based power generation, and (ii) the substitution of electricity imports from Viet Nam with domestic generation. The project's economic internal rate of return is recalculated at 14.9%, which exceeds ADB's current social discount rate (9.0%) and the social discount rate during project appraisal (12.0%). It also exceeds the 10.0% threshold in the extended annual review report guidelines.

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ADB's additionality is rated *excellent*. Without ADB's involvement, the project might not have been implemented. ADB's deal team closely followed the project's development from inception and was closely involved in structuring the loan and project documents. ADB provided a single-window approach to mobilize the entire project debt, mitigating any risks to the project of delayed financial closure. ADB secured concessional finance for the project, without which the sponsor's equity returns and the lender's debt serviceability requirement would not have been met. ADB's stamp of approval made the B lender comfortable with key project aspects regarding project viability, local law issues, due diligence requirements, and the Cambodian transfer and convertibility risk. This led to the B lender giving Sunseap a 14-year loan tenor, which until then was unavailable from Cambodian commercial banks.

ADB's investment profitability is rated *satisfactory*. The interest margin and fees on the ADB loan yield a net positive gross contribution to ADB.

ADB's overall work quality is rated *satisfactory*. ADB followed the project's progress from an early stage. The deal team worked intensively to set the commercial template and project structure that was replicated in subsequent solar projects in Cambodia. The loan was processed in a relatively short span of time, while ensuring adequate oversight. Despite some shortcomings during monitoring, the deal teams closely followed the project's performance and promptly approved consent requests. The portfolio monitoring teams spent a significant amount of time guiding a client unfamiliar with project finance and providing assistance in meeting technical, financial, safeguard requirements.

Finally, the project offered several lessons for ADB projects, focusing on scaling up renewable energy adoption in Asia's developing member countries. It showcased the role of concessional finance in promoting renewables in a frontier market. It highlights the need for greater scrutiny of sponsor's capacity (at mid-management and operating levels) to meet ADB's legal obligations. The review also suggests need to develop a simpler set of project finance documents, which can be readily understood (and adhered) to by less sophisticated clients. Finally, the review provides two lessons to improve project due-diligence, this relates to clearer identification of conflict of interests (among project participants) and the importance of systematically tracking project risks throughout the project processing cycle.

I. THE PROJECT

A. Project Background

1. Cambodia, a post-conflict country, recorded strong economic growth in the last two decades (with GDP growth averaging 7.9% p.a. between 1997-17).[1] This has led to Cambodia's transformation from among the poorest countries in the world to a lower middle-income country (with per capita GDP of $1,427 in 2017). Despite its economic achievements, Cambodia remained a largely rural[2] economy with a large section of population lacking access to electricity.[3] Broader access to electricity remained constrained due to unaffordable tariffs, a fragmented grid and limited institutional resources (financial and human) at Electricité du Cambodge (EDC) to push grid connectivity to rural areas.

2. The factors which contributed to this situation include: (i) the system's heavy dependence on hydropower (impacted by heavy seasonality) and expensive (and entirely imported) fossil fuel based generation. These resulted in Cambodian electricity tariffs being among the highest in Asia – with average prices at $0.30 per kWh versus $0.08 per kWh in China; and (ii) inadequate surplus left with EDC to make investments to the country's transmission and distribution networks (particularly in rural areas).

3. Asian Development Bank's (ADB) financial support to the project was expected to overcome these obstacles by (i) helping establish an off-grid, renewable energy solution in a disconnected region of Cambodia; (ii) reducing Cambodia's exposure to international fuel prices and EDC's electricity generation costs; and (iii) reducing first-mover barriers for private developers. While the cost of solar photovoltaic had reached grid parity across parts of Asia, solar photovoltaic had still not been adopted in Cambodia. ADB support aimed to overcome this obstacle by (i) establishing a precedent transaction with a bankable structure and project documents, such as power purchase agreement (PPA), land lease, interconnection agreement, and financing documents; and (ii) demonstrating that Cambodian independent power producers can access long-tenor project finance.

4. Given this backdrop, EDC initiated a tender to develop Cambodia's first utility-scale, 10-megawatt (MW) solar power plant. In February 2016, five bidders from Thailand, France, and Singapore participated in the process, which culminated in EDC awarding the project to Sunseap Asset (Cambodia) Co. Ltd.

B. Key Project Features

5. **Project overview.** The project included a 10 MW solar power plant (awarded on a build–own–operate basis) near Bavet, a city in Svay Rieng Province that is 150.0 kilometers (km) southeast of the capital city Phnom Penh; a 5.5 km transmission line connecting the plant to the Chrak Mates substation; and associated project facilities, such as a supervisory control and data acquisition system. The project's location was considered appropriate given the adequate solar

[1] Country Economic Indicators (accessible from the list of linked documents in Appendix 2). Manila; and ADB. 2018. *Basic Statistics 2018*. Manila.

[2] In 2017, 77.0% of Cambodia's population lived in rural areas, down from 80.8% in 2005 (ADB. 2018. *Basic Statistics 2018.* Manila; and ADB. 2018. *Key Indicators for Asia and the Pacific 2018.* Manila).

[3] In 2017, 31% of households lacked access to grid-connected electricity. Nearly 5 million Cambodians had no access to grid electricity and relied on car batteries, wood, and other traditional fuels for energy (Electricity Authority of Cambodia. 2018. *Salient Features of Power Development in Kingdom of Cambodia.* Phnom Penh).

resource,[4] reasonable land costs, and proximity to a load bearing center (Bavet).[5] The project represented an innovative solution for Svay Rieng, which had seen rising electricity demand from local industry but was still disconnected from the national grid and dependent on imports from Viet Nam. The project was expected to provide balancing power during the region's summer dry season (i.e., a period of peak demand with low generation from base hydropower plants) to curb rolling brownouts.

6. **Sponsors.** Sunseap is a special-purpose company that is owned 51% by Sunseap International Pte. Ltd., a subsidiary of Sunseap Group Pte. Ltd., a Singaporean solar developer; and 49% by SchneiTec Co. Ltd., a Cambodian solar equipment distributor.

7. **Engineering, procurement, and construction and operation and maintenance contract.** Schneitec Engineering Co. Ltd. implemented the project under a fixed-price, date-certain turnkey engineering, procurement, and construction (EPC) contract worth $11.6 million. EDC assigned both EPC and operation and maintenance (O&M) to Schneitec group companies.

8. **Power purchase agreement.** In August 2016, EDC, as the offtaker, entered into a 20-year PPA with Sunseap. The PPA had the following notable features.

(i) The PPA had a 20-year term from commissioning, a take-and-pay commitment of up to 16 gigawatt-hours (GWh), and a fixed tariff of $0.091 per kWh.[6] The tariff was "discovered" through a competitive tender and was below EDC's average tariff.
(ii) The tariff was invoiced in United States (US) dollars.
(iii) In case of termination because of a prolonged natural or political force majeure or an EDC risk event, Sunseap would be paid in US dollars the sum of (a) the total outstanding loans under the facility agreement net of debt service reserve account (DSRA) amount and insurance proceeds, and (b) the initial equity invested.
(iv) A project event of default gave EDC the option to purchase the facility at the greater of (a) the total outstanding loans under the facility agreement net of DSRA amount and insurance proceeds, or (b) 70% of the book value of the project assets net of insurance proceeds. Non-Cambodian political events[7] and natural force majeure events, such as epidemics and disasters caused by natural hazards, would result in payments only for electricity delivered. Sunseap should have insured against certain natural force majeure events, such as floods, fires, and earthquakes.
(v) Cambodian law was the governing law, with dispute resolution through arbitration in Singapore.
(vi) No Cambodian government guarantees were provided to backstop the PPA commitments.

9. ADB's actual financial assistance comprised (i) an A loan of $3.25 million, (ii) a B loan of $2.70 million, and (iii) a $3.25 million concessional loan from the Canadian Climate Fund for the Private Sector in Asia (CFPS) under the Clean Energy Financing Partnership Facility. The A loan and the CFPS loan had an 18-year tenor, while the B loan had a 14-year tenor.

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[4] The project's global horizontal irradiance of 1,907 kWh per square meter was among the highest in Southeast Asia.
[5] Bavet had a favorable demand–supply situation, registering strong economic growth after two special economic zones (SEZ) and gaming resorts were set up. Further, electricity demand was projected to increase from 14.4 MW in 2015 to 37.1 MW by 2024. Such demand could not be met by electricity imports from Viet Nam because the substation servicing the province was becoming increasingly congested.
[6] For generation beyond 16 GWh, the tariff payable by EDC goes down to $0.0546 per kWh.
[7] Acts of war and strikes outside Cambodia.

C. Progress Highlights

10. **Project development and construction.** Project bidding was started in February 2016. The tender attracted national and international bidders as 35 companies collected the bid documents and five companies submitted their proposals by March 2016, of which three companies were technically qualified. The final proposals of the three qualified companies were opened in May 2016 for evaluation in which Sunseap won the bid with the lowest tariff rate of $0.091 per kWh and was awarded the PPA in August 2016. Construction began in September 2016 with a scheduled commercial operations date (COD) of 29 August 2017. The project was physically completed on 30 August 2017, but COD was declared on 1 October 2017, 32 days after the scheduled COD, at EDC's request to monitor dispatch stability. This delay reduced the first month's revenue because of the lower pre-COD tariff under the PPA. However, EDC waived the penalties from the delayed COD. The project was completed slightly above budget; the actual project cost is $13.3 million, 0.7% higher than the original budget of $13.2 million.

11. **Operation.** From COD to December 2019, the project generated 35.3 GWh, higher than the P90 (base case) estimate. The plant's operational performance was satisfactory, with an availability factor of 99.0%. However, the plant's weather-adjusted performance ratio at EDC's meter of the project until September 2019 is about 2.6% lower than the P90 performance ratio estimated at financial close.[8]

II. EVALUATION

A. Project Rationale and Objectives

12. The project's rationale was well aligned with the government's "Rectangular Strategy" for Growth, Employment, Equity and Efficiency Phase III, which emphasized (i) increasing EDC's power generation capacity to meet rising demand; (ii) diversifying the country's sources of supply, away from hydropower and imported fossil fuels to domestic, renewable sources.[9] This would contribute toward meeting Cambodian Nationally Determined Contributions and reducing system-wide generation costs, a key impediment to wider access to electricity. The project was also well aligned with (i) ADB's country partnership strategy, 2014–2018 for Cambodia, which emphasized environmentally sustainable development;[10] and (ii) ADB's Energy Policy,[11] which focused on private sector-led solutions to meet Asia and the Pacific's rising energy demands.

B. Development Results

1. Contributions to Private Sector Development and ADB's Strategic Development Objectives

13. The contribution to private sector development is rated *excellent*. The project met most of its design and monitoring framework targets (Table 1). Despite being a small project with limited

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[8] The lender's technical advisor (LTA) attributed this to misalignment of photovoltaic modules, uncalibrated pyranometers, and supervisory control and data acquisition errors, all of which were later rectified.

[9] Government of Cambodia. 2013. *"Rectangular Strategy" for Growth, Employment, Equity and Efficiency Phase III*. Phnom Penh.

[10] ADB. 2014. *Country Partnership Strategy: Cambodia, 2014-2018*. Manila.

[11] ADB. 2009. *Energy Policy.* Manila.

resources committed by ADB, it created positive impacts on private sector development and ADB's strategic development objectives (para 14.).

Table 1: Design and Monitoring Framework Indicators, Targets, and Actual Achievement

High-Level Objectives	Count	Project Indicator	Target Value	Target Year	Description
Increased renewable energy capacity sustainably operated by the private sector	1	Annual energy generated	14 GWh per year	2018	Achieved. Annual generation post-COD exceeded 14 GWh.
	2	Reduced greenhouse gas emission	9,500 tCO_2 per year	2017–2027	Not achieved. Sunseap reported 8,250 tCO_2 of reduced emissions during 2019.
	3	Additional jobs created (operation)	10 workers	2018	Achieved. The project employs 10 full-time workers.
	4	Direct contribution (corporate tax) to the government	CONFIDENTIAL INFORMATION DELETED	2023	Achieved.
Power plant installed	5	Generation capacity built	10 MW	2017	Achieved at project COD.
	6	Transmission line connected to substation	5.5 km	2017	Achieved at project COD.
Local jobs generated	7	Additional jobs created (construction)	90 workers	2017	Achieved. The project created about 146 full-time equivalent jobs during construction.
	8	Local purchase of goods and services	$1.00 million	2017	Achieved. The project spent $1.36 million on local purchase of goods and services during construction.

COD = commercial operations date, GWh = gigawatt-hour, km = kilometer, MW = megawatt, tCO_2 = ton of carbon dioxide.
Source: Borrower submissions.

14. **Private sector development.** Prior to ADB's investment, there was no utility-scale solar plant in Cambodia. This was despite Cambodia having the highest solar potential[12] in the Greater Mekong region, which gave the country a technical solar potential of 8,000 MW. The project's implementation had the following impacts on private participation in Cambodia's energy sector.

(i) It showcased that solar photovoltaic had reached grid parity in Cambodia and that solar photovoltaic projects can be scaled up without government support.[13]

(ii) It improved the reliability of Bavet's power supply, a problem created by imbalances in the region's power supply.[14] Before the project, power demand in Bavet was about 40 MW, of which 20 MW was imported from Viet Nam. Irregular power supply during the summer months and expensive power were common complaints[15] at the nearby Manhattan special economic zone (SEZ)—the largest

[12] ADB. 2018. *Cambodia: Energy Sector Assessment, Strategy, and Road Map*. Manila. Cambodia had a global horizontal irradiance of 1,400–1,800 kWh per square meter per year.

[13] IJGlobal, 2019 *"News of the Project's successful commercial operation resulted in an increase an interest in Cambodia by developers previously wary of its high level of political risks"*; and submission of unsolicited bids.

[14] The imbalances in power supply between the dry and wet seasons were a result of high demand and low generation during the summer season, resulting in frequent brownouts. The problem was aggravated by the system's reliance on hydropower (52% of Cambodia's power supply), which typically runs at low capacity during the summer.

[15] Cambodian SEZs routinely cited electricity costs and supply as major complaints. In the Phnom Penh SEZ, electricity costs $0.20 per kWh, compared with $0.07 per kWh in Thailand and Viet Nam. Electricity accounted for about 25% of variable costs, depending on the industry. Frequent interruptions to electricity supply required firms to install

SEZ in Cambodia with more than 25,000 workers—and two other SEZs with 8,000 workers.[16] The resulting summer brownouts and work stoppage impacted the productivity of neighboring communities and factories. The project's implementation materially benefited the power situation in Bavet and contributed to displacing electricity imports from Viet Nam.

(iii) It helped prototype contractual arrangements for private investment in Cambodia's grid-tied solar projects. It also helped pilot-test the institutional and contractual arrangements, helping EDC and sector stakeholders familiarize themselves with operating a utility-scale solar photovoltaic project and giving them the confidence to tender four solar photovoltaic projects with 140 MW in installed capacity since 2017.[17] Feedback from ADB's Office of Public–Private Partnership suggested that the project's experience was instrumental in clarifying key regulatory risk and contractual issues during the structuring of the 100 MW Cambodia National Solar Park Project, the first 60 MW of which set a record low tariff of $0.03877 in Southeast Asia, helping ADB achieve its goal of providing affordable, sustainable energy to Cambodia's consumers.[18]

(iv) It demonstrated that long-term project finance could be raised at reasonable interest rates in Cambodia. Cambodia severely lacked project finance because of local banks' inability to offer tenors beyond 5–10 years and lack of credit appraisal skills for renewable projects, and lack of foreign bank presence. The project overcame this and raised a 14-year B loan from a commercial lender (BRED Bank) that then became comfortable in taking Cambodian risk if political risk, dispute settlement, and force majeure risk were mitigated under ADB's preferred creditor status.

(v) Building on the experiences gained with the project, (i) SchneiTec Co. Ltd. announced plans to implement a further 140 MW in Cambodia from 2018 to 2021; and (ii) Sunseap has implemented a further 300 MW in solar projects across Southeast Asia.

2. Economic Performance

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16. CONFIDENTIAL INFORMATION DELETED.

3. Environment, Social, Health, and Safety Performance

17. CONFIDENTIAL INFORMATION DELETED.

18. The project was classified as *category B* for the environment, *category C* for involuntary resettlement, and *category C* for indigenous peoples following ADB's Safeguard Policy Statement (2009). The project's potential environmental and social impacts were identified and measures to avoid, minimize, mitigate, and compensate for the adverse impacts were incorporated in the safeguard reports and plans. The scope of work for ADB's technical advisor during due diligence

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expensive backup generators using diesel power, costing about $0.28 per kWh (P. Warr and J. Menon. 2015. Cambodia's Special Economic Zones. *ADB Economics Working Paper Series*. No. 459. Manila: ADB).

[16] P. Warr and J. Menon. 2015. Cambodia's Special Economic Zones. *ADB Economics Working Paper Series*. No. 459. Manila: ADB.

[17] V. Petrova. 2020. Cambodia's Grid-Connected PV Capacity Grows to 150 MW — Report. *Renewables Now*. 13 April.

[18] ADB. 2019. ADB-Supported Solar Project in Cambodia Achieves Lowest-Ever Tariff in ASEAN. News release. 5 September.

included consideration of safeguards and social matters. An initial environmental and social examination per ADB's Safeguard Policy Statement requirements was prepared for the project by an experienced and qualified consulting company, and it included an ESMP that was to be fully implemented by Sunseap.

19. The project did not cause involuntary displacement impacts or impacts on vulnerable indigenous peoples. The entire land for the project site was purchased by an agent from 27 sellers through negotiation on a willing-buyer, willing-seller basis. An overhead transmission line for electricity evacuation used an existing canal and roadside right-of-way, which EDC selected to avoid involuntary displacement impacts. Sunseap did not undertake ADB's recommended action to monitor the transmission line construction for involuntary displacement impacts and work with EDC's construction subcontractor to mitigate, manage, and compensate any adverse impacts.

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4. Business Success

21. The project was completed largely on time and slightly above budget. It has since performed in a stable manner since COD, with higher than expected irradiance and availability levels (consistently above 99.0%), resulting in generation above P90 (and closer to P50 in 2019). However, business success is rated *less than satisfactory* because of the following factors.

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(ii) CONFIDENTIAL INFORMATION DELETED.
(iii) CONFIDENTIAL INFORMATION DELETED.

5. Overall Development Results Rating

22. The overall development outcome of this project is rated *satisfactory*.[19] The project demonstrated strong development impacts on private sector development and economic benefits to the host country, compensating for minor negatives associated with safeguard implementation and low financial returns for the sponsors (a known issue even during appraisal).

C. ADB Additionality

23. ADB's additionality was rated *excellent*. Before ADB's investment, Cambodia did not have utility-scale solar projects. Project finance was scarce, and investors were wary of underwriting political risk associated with investing in Cambodia. Given such context, ADB brought significant financial and nonfinancial additionality to this project.

24. **Financial additionality.** ADB's team provided EPC and O&M inputs to the project and loan documents to make this a bankable transaction. It provided a single-window approach by arranging the entire project debt, mitigating risks of a delayed financial close. ADB provided an 18-year loan, which was otherwise difficult to obtain or not available from Cambodia's commercial banks; and syndicated $3.0 million in commercial finance through a participating B loan.[20] It also brought $3.25 million in concessional financing. The amount of concessional finance was

[19] The overall development outcome rating is based on a synthesis of the project's (i) contributions to private sector development and ADB strategic development objectives; (ii) economic performance; (iii) environment, social, health, and safety performance, and (iv) business success.

[20] While ADB remained the lender of record to Sunseap, BRED Bank took the same security and credit risk as ADB (but shorter duration risk).

justified, as it allowed the project to attain minimum debt service cover of CONFIDENTIAL INFORMATION DELETED and CONFIDENTIAL INFORMATION DELETED equity internal rate of return at appraisal. Without concessional financing, the project would have become unviable for the sponsors and lenders, as the project equity internal rate of return would have declined to CONFIDENTIAL INFORMATION DELETED, and the minimum debt service coverage ratio would have declined to CONFIDENTIAL INFORMATION DELETED.

25. **Nonfinancial additionality.** ADB had the following key nonfinancial additionalities:

(i) ADB structured and underwrote the entire project debt, which helped it build credibility with the Sunseap Group. ADB expects to deepen its engagement with the Sunseap Group, which is rapidly becoming a leading renewable energy developer in Southeast Asia and can substantially help ADB meet its climate change objectives.

(ii) ADB's environmental and safety diligence identified the key environmental impacts and mitigants. The safeguard appraisal, although not required under local law, demonstrated good practice in conducting an initial environmental examination and drafting an environmental management plan. ADB's reviews ensured fair treatment to land sellers in the neighboring community.

(iii) ADB's presence as the lead bank granted an implicit stamp of approval to the transaction and gave BRED Bank the comfort that Sunseap's safeguard, commercial, and integrity due diligence would meet international standards. BRED Bank also benefited from ADB's preferred creditor status, which exempted its interest payments from withholding tax and offered protection from transfer and convertibility risk. In return, ADB expected BRED Bank to gain confidence from this transaction and be able to pursue renewable project financing independently in Cambodia.

(iv) CONFIDENTIAL INFORMATION DELETED.

D. ADB Investment Profitability

26. ADB's investment profitability is rated *satisfactory*. ADB's US dollar-denominated loan, which included a step-up interest rate, generated a weighted average margin of CONFIDENTIAL INFORMATION DELETED. The margin was sufficient from a risk–return perspective, as CONFIDENTIAL INFORMATION DELETED.[21] Sunseap has been compliant with its debt service obligations.

E. ADB Work Quality

27. ADB's overall work quality is rated as *satisfactory*, despite some areas of improvement identified in paras. 32–34.

28. Screening, appraisal, and structuring are rated *satisfactory*. ADB's deal team closely followed the project from inception to bidding. Its upstream engagement with bidders gave it the confidence to reach a quick financial close and eventually contributed to the success of the

[21] ADB. Office of Risk Management 2016. *Final Credit Review Memo for Cambodia Solar Power Project*. Manila. (Internal). CONFIDENTIAL INFORMATION DELETED.

efficient and transparent bid process, resulting in a competitive tariff for consumers. After the project award, ADB's deal team efficiently completed the diligence and structuring process.[22]

29. ADB's appraisal of the project covered the usual areas of technical, legal, environment, social, and financial diligence and development assessment to ensure a fundamentally sound project. The project was well structured, with prudent financial covenants (minimum debt service coverage ratio of 1.20x at P90) and a cash flow waterfall mechanism with onshore and offshore security, which strengthened the lender's monitoring and ability to exercise control during supervision. In addition, the safeguard team should be commended for its efforts to build capacity and transfer safeguard knowledge to the Sunseap Group. This helped the Sunseap Group develop a framework to identify, assess, and manage environment and social concerns at both the organization and the project.

30. Monitoring and supervision are rated as *satisfactory*. Despite being a simple and relatively small project, it had the contractual complexities of a larger solar project. ADB closely monitored Sunseap's performance through client and LTA monitoring reports and promptly processed consent requests CONFIDENTIAL INFORMATION DELETED.

F. Overall Evaluation

31. The project's overall evaluation is considered *successful* (Table 2). The project is still highly relevant for Cambodia, contributing to Bavet's sustainable economic growth and delivering clean power.

Table 2: Evaluation Summary

Indicator	Unsatisfactory	Less than Satisfactory	Satisfactory	Excellent
A. Development Results			✓	
(i) Contributions to private sector development and ADB strategic development objectives				✓
(ii) Economic performance			✓	
B. ADB Additionality				✓
C. ADB Investment Profitability			✓	
D. ADB Work Quality			✓	
(i) Screening, appraisal, and structuring			✓	
(ii) Monitoring and supervision			✓	
E. Overall Rating	**Unsuccessful**	**Less than Successful**	**Successful**	**Highly Successful**
			✓	

ADB = Asian Development Bank.

III. ISSUES, LESSONS, AND RECOMMENDED FOLLOW-UP ACTIONS

A. Issues and Lessons

32. **Role of concessional finance in market development.** The project demonstrates the role played by concessional finance in making a project viable in a frontier market. It showed how financiers can use a small project to test the robustness of the contractual arrangements in a new

[22] ADB completed concept clearance in July 2016, received Board approval in December 2016, signed the loan document in April 2017, and made the first disbursement in May 2017. This aspect is noteworthy as the transaction was in a new jurisdiction for ADB's Private Sector Operations Department, was with a first-time borrower, and involved multiple parties to reach financial close (i.e., syndications and concessional finance committee).

market and generate lessons for projects and developers. Based on government information, the frequency of queries and unsolicited proposals for developing solar projects increased after the project was commissioned. Future project designs, especially projects with concessional finance, should incorporate a formal knowledge sharing component that would systematically and widely disseminate the project's lessons to a wider investor and developer community.[23]

33. **Due diligence improvements.** This review also offers some lessons on diligence for future projects.

(i) The sponsors played multiple roles in Sunseap: equity contributors, EPC contractor, operator, and land lessor. This integrated approach is positive as it provides strong oversight by the owners and risk mitigation in frontier markets where hiring quality contractors is an issue. However, it also potentially creates conflicts of interests, which should be closely scrutinized by benchmarking project costs, operator's services, and terms with market prices and other ADB projects at appraisal; adequately documented; and contractually mitigated with independent reviews before terms are modified.[24]

(ii) ADB's workflows make it difficult to track the risks, diligence issues, and mitigants raised during project approval.[25] This issue can get compounded in cases when there is frequent staff turnover at ADB or Sunseap, resulting in a loss of institutional knowledge. During monitoring, ADB incorporated several improvements in its own systems for tracking items such as project security. This review suggests further improvements in information technology systems to (a) systematically track technical and legal risks identified during the reviews and how they were mitigated contractually during the approval process; and (b) ensure all requested diligence items during concept or final review are clearly covered in the technical, legal, and other consultants' reports; and filed separately by the deal teams.

B. Recommendations and Follow-Up Actions

34. Monitoring this transaction took up considerable internal ADB resources relative to its size. In future deals of this size, ADB should think about how it can (i) limit the monitoring burden, such as short-form financing documentation; low-maintenance structuring (e.g., cash DSRA); and truncated, templated reporting; CONFIDENTIAL INFORMATION DELETED. ADB will continue to monitor the financial, operational, and safeguard performance of the project until maturity.

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[23] This knowledge sharing component should go beyond brief case studies and cover (i) details about the key design and implementation issues faced by a photovoltaic project developer in that frontier market, such as accuracy of insolation data, outline of PPA terms, permitting requirements (and issues faced), site selection considerations, module efficiency under ambient conditions, insurance costs, and actual O&M costs; and (ii) an overview of the utility demand profile and how it matches the power generation profile from the solar plant.

[24] The LTA and EPC contract reviews noted that multiple terms in the EPC and O&M contracts deviated from market terms and favored the contractors. For instance, the amount for delay liquidated damages, capital expenditure contingency, annual land lease, and annual operating expenses were all below market rates at the time.

[25] Without qualifying the quality of diligence, this review noted missing items from the LTA diligence: (i) a detailed offtaker analysis, which required a long-term credit assessment of EDC, the impact of Cambodia's rapidly expanding power development plan on EDC's financials, fiscal analysis of the Cambodian government and its ability to support EDC; (ii) the adequacy of EPC agreements, particularly liquidated damages available for performance shortfall by the plant (due to design defects); (iii) an overview of Cambodian power regulations; and (iii) an update to the registration of the project land lease agreement and the conversion of land use from agricultural to commercial or industrial, which was not available.

PROJECT-RELATED DATA

Table A1.1: Investment Identification

Item	Description
1. Country	Cambodia
2. Project Number	50248-001
3. Loan Number	3495
4. Type of Business	Renewable energy generation—solar
5. Project Title	Cambodia Solar Power Project
6. Investee Company and/or Borrower	Sunseap Asset (Cambodia) Co. Ltd.
7. Amount of ADB Assistance	ADB direct loan: $3.25 million B loan: $2.70 million CFPS: $3.25 million

ADB = Asian Development Bank, CFPS = Canadian Climate Fund for the Private Sector in Asia.
Source: Asian Development Bank.

Table A1.2: Investment Data

Item	Description
1. Concept Clearance Approval	25 July 2016
2. Date of Board Approval	7 December 2016
3. Signing Date of Legal Agreements	4 April 2017
4. Terms of Loans	• CONFIDENTIAL INFORMATION DELETED. • CONFIDENTIAL INFORMATION DELETED. • CONFIDENTIAL INFORMATION DELETED. • CONFIDENTIAL INFORMATION DELETED. • CONFIDENTIAL INFORMATION DELETED.
5. Loan Amount and Date of Initial Disbursement	OCR: $3.25 million, 17 May 2017 B Loan: $2.70 million, 17 May 2017 CFPS: $3.25 million, 17 May 2017
6. Repayment	OCR and CFPS First repayment date: 30 April 2018 Final repayment date: 31 October 2034 B Loan First repayment date: 30 April 2018 Final repayment date: 31 October 2030

CFPS = Canadian Climate Fund for the Private Sector in Asia, OCR = ordinary capital resources.
Source: Asian Development Bank.

Table A1.3: Summary of Project Cost and Funding Sources

Particulars	RRP Amount ($ million)	Actual Amount ($ million)	Share of Total (%)
A. Project Implementation Cost	13.2	13.3	100
Total project cost	**13.2**	**13.3**	**100**
B. Sources of Funds			
1. Equity	3.4	4.1	31
2. Debt			
a. ADB direct loan	3.6	3.2	24
b. B Loan	3.0	2.7	20
c. Concessional finance	3.2	3.2	25
Total Debt	**9.8**	**9.2**	**69**
Total sources of funds	**13.2**	**13.3**	**100**

ADB = Asian Development Bank; RRP = report and recommendation of the President.
Note: Numbers may not sum precisely because of rounding.
Source: ADB. 2016. *Report and Recommendation of the President to the Board of Directors: Proposed Loans and Administration of Loan to Sunseap Asset (Cambodia) Co. Ltd. for the Cambodia Solar Project in Cambodia*. Manila.

RESULTS AND RATINGS FOR PROJECT CONTRIBUTIONS TO PRIVATE SECTOR DEVELOPMENT AND ADB STRATEGIC DEVELOPMENT OBJECTIVES—INFRASTRUCTURE

Results Area	Actual Achievements	Rating	Justification	Potential Future Achievements	Risk
1. Within company PSD effects					
1.1 Improved skills. New or strengthened strategic, managerial, operational, technical, or financial skills.	The project was the first time Schneitec implemented a grid-tied solar photovoltaic project. Based on its experience in bidding, developing, and managing the Bavet project, Schneitec secured two more solar photovoltaic projects in Cambodia (a 60 MW project in Kampong Speu and a 20 MW project in Bavet). These projects were completed in 2019 and 2020 and drew upon the lessons of the 10 MW Bavet project.	Satisfactory		Schneitec and Sunseap continue to be active in bidding for new projects in Southeast Asia. They are expected to improve internal capabilities in project execution and scale in the near term.	Low
1.2 Improved business operations. Improved ways to operate the business and compete, as seen in investee operational performance against relevant best industry benchmarks or standards.					
1.3 **Innovation**. New or improved infrastructure design, technology, and service delivery; ways to cover or contain costs, manage demand, or optimize utilization; improved risk allocation between private companies and government; financial structure, etc.	Before ADB's investment, there was no utility-scale solar plant in Cambodia. The project's successful bid and implementation demonstrate that solar photovoltaic had reached grid parity in Cambodia. It would therefore not need any further government support to scale-up.	Excellent		The project's implementation has had a perceptible demonstration effect and created increased investor interest to invest in Cambodia's solar photovoltaic sector.	Low
1.4 **Catalytic element.** Mobilizing or inducing more local or foreign market financing or foreign direct investment in the company.	The project mobilized its entire project debt through ADB, which comprised (i) ADB's OCR loan of $3.25 million with an 18-year tenor, (ii) a B loan of $2.70 million, and (iii) concessional finance of $3.25 million.	Satisfactory			Low
2. Beyond company PSD effects					
2.1 Private sector expansion.	The project's implementation resulted in	Excellent			

Results Area	Actual Achievements	Rating	Justification	Potential Future Achievements	Risk
Contribution by a pioneering or low-profile project that facilitates in its own right, or paves the way for, more private participation in the sector and economy at large.	greater crowding in of international investors into Cambodia's solar power sector. The project's ongoing operations contributed to reliable power supply in Bavet, thereby alleviating brownouts and work stoppage at nearby SEZs. The project's implementation also gave confidence to EDC and the Cambodian government to bid out more projects through competitive auction. The project confirmed EDC's belief that it could benefit from falling photovoltaic prices to reduce electricity costs for consumers.				
2.2 Competition. Contribution of new competition pressure on public and other sector players to raise efficiency and improve access and service levels in the industry.					
2.3 Demonstration effects. Adoption of new skills, improved infrastructure assets and services, more efficient processes, maintenance regimes, improved standards, risk allocation and mitigation beyond the project company.	Feedback from ADB's Office of Public–Private Partnership suggests that the project's experience was instrumental in clarifying key risk and contractual issues during the structuring of the 100 MW Cambodia National Solar Park Project, which set a record low tariff of $0.03877 in the GMS, helping ADB achieve its goal of providing affordable, sustainable energy to Cambodia's consumers.	Excellent		Based on the project's experience, EDC is expected to continue to use competitive auctions to reduce supply costs and increasingly rely on the private sector to develop utility-scale solar projects.	Low
2.4 Linkages. Relative to investments, the project contributes notable upstream or downstream linkage effects to business clients, consumers, suppliers, key industries, etc. in support of growth.	The project had positive downstream impacts as it improved the power situation in Bavet.	Satisfactory			

Results Area	Actual Achievements	Rating	Justification	Potential Future Achievements	Risk
2.5 Catalytic element. Mobilizing or inducing more local or foreign market financing or foreign direct investment in the sector (beyond the company) through pioneering or catalytic finance.	The project prototyped contractual arrangements for private investment in Cambodia's grid-tied solar projects. It helped EDC and sector stakeholders familiarize themselves with operating a utility-scale solar photovoltaic project, giving them the confidence to tender four solar photovoltaic projects with 140 MW in installed capacity since 2017. As evidence of increased investor interest in the sector, more than 100 bidders (mostly international players) participated at the bid participants conference for the 100 MW Solar park project. The competitive auction resulted in a record low tariff in the GMS.	Satisfactory		The project's successful financial close signaled to the investor community that long-term project and concessional finance could be raised for Cambodian projects. It also increased interest from commercial banks, particularly from the PRC, to invest in Cambodian projects.	Low
2.6. Affected laws, frameworks, and regulation. Contributes to improved laws and sector regulation for PPPs, concessions, joint ventures, and build–operate–transfer projects; and liberalizing markets as applicable for improved sector efficiency.					
3. Contribution to other ADB strategic objectives					
3.1 Sector development outputs. Contribution to other sector development outputs not captured under point 2, such as capacity or network expansion.	The project achieved most DMF targets on increased renewable energy generation, reduced GHG emissions, job creation, and local purchase of goods (main text, Table 1). The project also contributed to ADB's country partnership strategy and Energy Policy by (i) helping establish an off-grid, renewable energy solution in a disconnected region of Cambodia; (ii)	Excellent			

Results Area	Actual Achievements	Rating	Justification	Potential Future Achievements	Risk
	reducing Cambodia's exposure to international fuel prices and EDC's electricity generation costs; (iii) reducing first-mover barriers for private developers; and (iv) demonstrating that Cambodian IPPs can access long-tenor project finance.				
3.2 Sector development outcomes. Contribution to other sector development outcomes not captured under point 2, such as increased infrastructure utilization or consumption, improved in-country connectivity, and improved energy security.	Feedback from ADB's Office of Public–Private Partnership indicated that the project's experience was instrumental in clarifying key risk and contractual issues during the structuring of the 100 MW Cambodia National Solar Park Project.	Excellent		SERD is working with the Government of Cambodia to prepare a solar road map, mapping the solar potential and investment plan over a long term. This is expected to significantly increase solar penetration in Cambodia.	Low
3.3 Inclusion. Improved access to, availability, or affordability of infrastructure services for the poor and other disadvantaged groups.	At the time of project bid, the project's tariff was significantly lower than EDC's average tariff. This improved EDC's financial position and contributed toward its goal of making electricity more accessible to average Cambodians.	Satisfactory			
3.4 Job creation. Creation of additional sustainable jobs or self-employment. Distinguish between jobs created within and beyond the company.	The project created 146 local jobs during construction, significantly higher than the 90 expected at appraisal. The project continues to employ 10 people during operation.	Excellent			
3.5 **Regional integration.** Project contributions to regional cooperation and integration by facilitating trade, cross-border mobility, cross-border power supplies, etc.					
4. Overall Rating		**Excellent**			

ADB = Asian Development Bank; DMF = design and monitoring framework; EDC = Electricité du Cambodge; EHS = environment, health, and safety; ESMP = environmental and social management plan; GHG = greenhouse gas; GMS = Greater Mekong Subregion; IPP = independent power producer; MW = megawatt; OCR = ordinary capital resources; PPP = public–private partnership; PRC = People's Republic of China; PSD = private sector development; Schneitec = Schneitec Engineering Co. Ltd.; SERD = Southeast Asia Department; SEZ = special economic zone.
Source: Borrower inputs, and ADB assessment.

SECTOR REVIEW

A. Sector Framework

1. Two main government agencies handle Cambodia's energy sector: the Ministry of Mines and Energy (MME) and the Electricity Authority of Cambodia (EAC). They were set up pursuant to the Electricity Law, 2001.

2. The MME is the main agency setting and administering government policies, strategies, and planning. It has three core departments: (i) the Department of Energy Development, which handles energy and electricity planning; (ii) the Department of Hydropower; and (iii) the Department of Technical Energy, which handles renewable energy (other than hydropower) and energy efficiency.

3. The EAC, the electricity regulator, is an autonomous agency that regulates electricity services, such as issuing rules, regulations, and procedures on power market operations; awarding licenses to power service suppliers; and setting tariffs. The agency monitors, guides, and coordinates operators in Cambodia's energy sector; and makes sure all operators follow the MME policies, guidelines, and technical standards.

4. Electricité du Cambodge (EDC), a state-owned vertically integrated utility owned by the MME and the Ministry of Economy and Finance, generates, transmits, and distributes electricity in areas assigned to it by the EAC. EDC purchases from independent power producers, most of which are joint ventures between Cambodian and foreign investors. EDC mainly supplies to Phnom Penh, the capital city, while rural areas are supplied by rural electricity enterprises (REEs), which operate as licensees that purchase power from EDC and sell to local distribution networks. REEs may have their own, normally diesel-powered generation assets.

5. The Ministry of Economy and Finance, as EDC co-owner, facilitates EDC's access to long-term and concessional finance. The Ministry of Environment reviews and approves environmental assessments and environmental management plans for all energy projects.

6. To address the increased need for electricity supply to support economic growth, there has been a more rapid development of electricity generation capacity, mainly driven by the private sector. Private domestic and foreign investors are encouraged to invest in (i) oil and gas exploration and production, (ii) energy generation from various energy resources, and (iii) the high-voltage transmission system.

B. Electricity Demand

7. Electricity demand in Cambodia has been increasing by about 18% per year from 2010 to 2016. In 2018, per capita consumption 532 kilowatt-hours (kWh),[1] an almost fivefold increase from 112 kWh in 2008.[2] Consumption is forecast to grow 9% per year from 2015 to 2040, driven mainly by population increase, accelerated economic activity, and rural electrification.[3] In 2000, only 16% of Cambodia's population had access to power. In 2018, 87% of villages and 73% of households are connected to the grid.[4]

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1 Fitch Solutions. Cambodia Power Report Q1 2020 (December 2019).
2 Energypedia. Cambodia Energy Situation (accessed 27 April 2020).
3 Government of Cambodia, General Department of Energy, General Department of Petroleum, and MME. 2019. *Cambodia Basic Energy Plan*. Jakarta: Economic Research Institute for ASEAN and East Asia.
4 J. Ferrie. 2018. Cambodia to Boost Clean Energy Use — but Coal Plants Planned Too. *Reuters*. 7 November.

8. Residential consumers have historically been the largest electricity consumer (except in 2016), comprising 40% of total consumption in 2017, followed by commercial and public services at 37%, and industry at 23%.[5] Of the peak demand in 2017, 70% came from Phnom Penh, where about 13% of Cambodia's population lives and where businesses and industries are concentrated.[6]

C. Electricity Supply

9. Cambodia's installed capacity was at 2,650 megawatts (MW) in 2018, a 14% increase from 2017. Energy delivered reached 9,307 gigawatt-hours (GWh) in 2018, with 50% coming from hydropower, 20% from coal, 17% from imported electricity, 10% from fuel, and 3% from renewable energy. Of the 1,440 GWh in imported electricity, 65% came from Viet Nam and 29% came from Thailand.[7] While Cambodia's electricity demand used to be dependent on imports, domestic power sources now account for more than 80% of local consumption because of a recent surge in investments, with a declared goal of being self-sufficient in the medium term.[8]

D. Electricity Transmission

10. Cambodia has had a fully integrated high-voltage transmission system in place since 2015, but it has been relying on power from Viet Nam, Thailand, and the Lao People's Democratic Republic (Lao PDR) for certain areas. The national grid is interconnected to Viet Nam synchronously via a 230-kilovolt (kV) line in the southeast, to Thailand via a 115 kV line in the northwest, and to Lao PDR via a 115 kV line in the northeast. Additional medium-voltage supply is available in several areas along the border through 22 kV extension lines from these grids. Cambodia is working on improving supply quality by increasing the number of REE licensees connecting to the grid supply by expanding the high-voltage transmission system and the medium-voltage sub-transmission system. The national grid supplies electricity to 19 of the country's 25 provinces via 30 substations. The 2015 Power Development Plan (PDP) aims to connect the remaining provinces to the national grid by expanding the transmission network by about 2,600 kilometers.[9] This transmission grid development is also essential in creating an integrated power market in the Greater Mekong Subregion.[10]

E. Electricity Supply and Demand Forecast

11. In the PDP's base case projection, electricity demand would grow 8.8% per year to 2030. In the same period, energy demand is expected to rise to 18,000 GWh—a more than threefold increase from 2015. This translates to peak demand forecasts of 1,412 MW in 2020 and 3,256 MW in 2030. At 2,650 MW, actual installed capacity in 2018 already exceeded the PDP's projected installed capacity of 2,526 MW by 2020. The more aggressive Cambodia Basic Energy Plan (BEP) prepared in 2019 projected total installed capacity to increase to 4,856 MW by 2025 (3,811 MW in the PDP) and 7,081 MW by 2030 (4,716 MW in the PDP). Both the PDP and the

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[5] International Energy Agency. Data & Statistics (accessed 28 April 2020).

[6] Asian Development Bank. 2018. *Cambodia: Energy Sector Assessment, Strategy, and Road Map*. Manila.

[7] EDC. 2019. *Cambodia Power Situation*. Presentation prepared for the 25th Meeting of the Regional Power Trade Coordination Committee. Thailand. 21–22 March.

[8] European Chamber of Commerce in Cambodia. 2018. *Energy in Cambodia*. Bangkok: Netherlands Embassy in Bangkok.

[9] Chugoku Electric Power Co., Inc. 2015. *The Project on Revision of Cambodia Power Development Master Plan*. Presentation prepared for the Government of Cambodia. Cambodia. September.

[10] Asian Development Bank. 2019. *Report and Recommendation of the President to the Board of Directors: Proposed Loan and Administration of Loan, Grant, and Technical Assistance Grant to the Kingdom of Cambodia for the National Solar Park Project*. Manila.

BEP assume that imported supply will not be needed to meet the required electricity demand by 2025 onward (footnote 3). The projected power generation by fuel type in the PDP and the BEP is summarized in Table A3.1.

Table A3.1: Power Generation by Fuel Type (%)

	Power Development Plan 2015			Basic Energy Plan 2019		
Fuel Type	**2020**	**2025**	**2030**	**2020**	**2025**	**2030**
Coal and gas	52.2	64.8	67.5	56.0	46.4	35.0
Oil	15.0	9.2	7.2	8.1	0.0	0.0
Hydropower	32.7	22.7	18.8	35.1	46.8	55.0
Renewable energy (biomass)	0.0	3.2	5.0	0.0	4.3	6.5
Renewable energy (solar, wind)	0.1	0.1	1.6	0.7	2.2	3.5
Total	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**	**100.0**

Note: Percentages may not total 100% because of rounding.
Sources: Chugoku Electric Power Co., Inc. 2015. *The Project on Revision of Cambodia Power Development Master Plan*. Presentation prepared for the Government of Cambodia. Cambodia. September; and Government of Cambodia, General Department of Energy, General Department of Petroleum, and Ministry of Mines and Energy. 2019. *Cambodia Basic Energy Plan*. Jakarta: Economic Research Institute for ASEAN and East Asia.

12. To keep up with demand, Cambodia has increased its imports of coal and petroleum from other Association of Southeast Asian Nations members such as Indonesia and Singapore. This has affected Cambodian economic growth because the formula defining gross domestic product deducts imports. Curbing coal, oil, and electricity demand to the gross domestic product is key and has led to the MME's preparation of the comprehensive BEP, which will contribute to (i) saving conventional energy consumption, such as oil; and (ii) using domestic energy, such as hydropower and biomass. While the PDP projects that by 2030, 50% of generating capacity will come from coal and gas and 34% from hydropower, the BEP expects that 66% of generating capacity will be from hydropower and only 17% will come from coal and gas. The projected share of renewable energy also increased from 10% in the PDP to 16% in the BEP (Table A3.2).

Table A3.2: Revised 2030 Targets for Generation Capacity

Source	**PDP 2015** (MW)	**Share** (%)	**BEP 2019** (MW)	**Share** (%)	**Difference 2019–2015**
Coal and gas	2,373	50	1,230	17	(1,143)
Oil	251	5		0	(251)
Hydropower	1,602	34	4,694	66	3,092
Renewable energy[a]	490	10	1,157	16	667
Total	**4,716**	**100**	**7,081**	**100**	**2,365**

( ) = negative, BEP = Basic Energy Plan, MW = megawatt, PDP = Power Development Plan.
Note: Percentages may not total 100% because of rounding.
[a] Biomass, solar, and wind.
Source: Government of Cambodia, General Department of Energy, General Department of Petroleum, and Ministry of Mines and Energy. *Cambodia Basic Energy Plan*. 2019.

F. Tariff Structure

13. In Cambodia, charged electricity tariffs depend on the cost of the electricity generated or purchased. The largest components of this cost are (i) the cost of electricity from own generation, (ii) purchases from independent power producers, and (iii) purchases from neighboring countries or other licensees. Because of factors that raise generation costs—such as small generation systems running on imported diesel fuel, high distribution network power losses, and the lack of subsidies—Cambodia has one of the highest electricity tariffs among its neighboring countries

(footnote 3). Industrial tariffs in Cambodia from 2014 to 2015 are $0.17–$0.21 per kWh, whereas the highest for Lao PDR, Thailand, and Viet Nam is $0.07–$0.11 per kWh.

14. The difference between residential tariffs is even higher, ranging from $0.18 to $0.26 per kWh in Cambodia when the highest in the three neighboring countries is $0.12 per kWh.[11] To address this, the Cambodian government introduced a government subsidy program in 2015 to reduce the electricity tariff from 2016 to 2020 in areas supplied by the national grid. From 2016, the tariff for customers that consume less than 10 kWh per month is about $0.12 per kWh, while the tariff for customers that consume less than 50 kWh per month is about $0.15 per kWh. Further, Cambodia reduced its reliance on diesel, imports, and grid extension, which have lowered the cost of supply (footnote 10).

15. Electricity tariffs are also generally higher in rural areas than in urban areas. The development of a grid system helps stabilize the tariffs of licensees purchasing electricity from the grid, but many rural areas have yet to be reached by the national grid. Tariffs also decrease when more than 50% of the villages are covered by electricity-licensed areas. The higher the percentage of villages covered, the lower the applied tariffs. Tariffs for less than 10 kWh per month and 11–50 kWh per month have the largest gaps between urban and rural areas. In 2015, tariffs for customers that use less than 10 kWh per month are $0.05 per kWh higher in rural areas than in urban areas. The government's 2015 tariff reduction plan aims to drastically decrease this gap in 2020 for purchase type. However, Cambodia's tariff is still expected to be higher than the tariff in neighboring countries (footnote 3).

[11] United States Agency for International Development. 2015. *The Business Case for: Solar PV in Cambodia*. Washington, DC.

BORROWER'S FINANCIAL ANALYSIS

Table A4.1: Financial Statements (2017–2026)
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Table A4.2: Financial Statements (2027–2037)
CONFIDENTIAL INFORMATION DELETED.

REEVALUATION OF THE FINANCIAL INTERNAL RATE OF RETURN AND WEIGHTED AVERAGE COST OF CAPITAL

1. The project entailed the construction of a 10-megawatt solar power plant in Bavet City, Svay Rieng Province, about 150 kilometers from the capital Phnom Penh. It is Cambodia's first utility-scale solar power plant and first competitively tendered renewable energy independent power producer project.

2. This analysis covers 20 years from 2017 to 2037. All values are adjusted to reflect 2017 prices. Cambodia's inflation data are readily available from various public sources.

A. Financial Internal Rate of Return

3. CONFIDENTIAL INFORMATION DELETED.

Table A5.1: Real Financial Internal Rate of Return
($'000)
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B. Weighted Average Cost of Capital

4. CONFIDENTIAL INFORMATION DELETED.

Table A5.2: Weighted Average Cost of Capital
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C. Sensitivity Analysis

5. The FIRR's sensitivity to increased operating expenses and reduced electricity production was analyzed. Table A5.3 shows the base case and the effects of adverse scenarios.

Table A5.3: Sensitivity Analysis
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6. CONFIDENTIAL INFORMATION DELETED.

REEVALUATION OF THE ECONOMIC INTERNAL RATE OF RETURN

A. Overview

1. In 2016, the Asian Development Bank (ADB) approved a loan of up to $6.60 million and the administration of a loan of up to $3.25 million[1] to Sunseap Asset (Cambodia) Co. Ltd. for the 10-megawatt Cambodia Solar Power Project in Bavet City, Svay Rieng Province.[2] It was the country's first utility-scale solar plant and only solar park until 2018.[3] The project has a 20-year power purchase agreement with Electricité du Cambodge, a state-owned company. Construction started in July 2016, and the generation of electricity started in October 2017.

2. Cambodia's electrification rate of 56% in 2014 was among the lowest in the Southeast Asia region.[4] Coal and imported electricity accounted for more than 50% of electricity generation. During that year, demand for electricity was estimated to grow 9.4% per year until 2020, partly because of a shift in economic activities from agriculture to manufacturing and services.

3. Moving to renewable energy sources, such as solar, seemed promising as the country receives 1,450–1,950 kilowatt-hours (kWh) per square meter of solar irradiation yearly, translating to a total potential energy output of 12,000 gigawatt-hours (GWh) per year.[5] However, the high cost of solar panels before 2016 limited the development of solar plants and the use of solar home systems. Previous strategies focused on advancing baseload hydropower and coal plants. The country also lacked the resources needed to implement new renewable energy projects.

4. Bavet borders Moc Bai in Viet Nam, and Svay Rieng Province is home to special economic zones. During project planning, Bavet was largely dependent on electricity imports from Viet Nam. In the report and recommendation of the President (RRP), the project was expected to generate an annual average of 14.0 GWh based on a 16% expected capacity factor in a P90 scenario,[6] and this was estimated to meet 25% of the city's energy needs. Commercial operation started on 1 October 2017. The project delivered 3.4 GWh in October–December 2017, 15.7 GWh in 2018, and 16.1 GWh in January–September 2019. The actual output is 8% higher than the projected output in the P90 scenario.

B. Key Assumptions and Methodology

5. The economic reevaluation for the project followed ADB's Guidelines on the Economic Analysis of Projects[7] for 2017–2037 operation at 2017 prices. The economic analysis used the world price as numéraire, applying a standard conversion factor of 0.95 and a shadow wage rate factor of 0.8 for unskilled labor as in the original calculation. Nominal prices were converted to real values of the base year of 2017 using the cost escalation factor of the original financial model.

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[1] The loan was provided by the Canadian Climate Fund for the Private Sector in Asia under the Clean Energy Financing Partnership Facility.

[2] ADB. 2016. *Report and Recommendation of the President to the Board of Directors: Proposed Loans and Administration of Loan to Sunseap Asset (Cambodia) Co. Ltd. for the Cambodia Solar Project in Cambodia*. Manila.

[3] Open Development Cambodia. 2015. Renewable Energy Production. 7 September.

[4] Government of Cambodia, National Institute of Statistics and Directorate General for Health; and ICF International. 2015. *Cambodia Demographic and Health Survey 2014*. Phnom Penh, Cambodia; and Rockville, Maryland, United States.

[5] ADB. 2015. *Renewable Energy Developments and Potential in the Greater Mekong Subregion*. Manila.

[6] P90 denotes 90% likelihood that the predicted annual energy production or higher will be achieved.

[7] ADB. 2017. *Guidelines for the Economic Analysis of Projects*. Manila.

6. The following are the main differences between the ex ante analysis and this ex post analysis:

(i) The RRP used 12% as the social discount rate, but current ADB guidelines recommends a 9% social discount rate (footnote 7). The results using a 12% social discount rate are summarized in Table A6.2.

(ii) The P90 base case in the original economic analysis corresponded to an average annual power generation of 14 GWh or a capacity factor of 16%. The ex post analysis used 2017–2019 values in its projections, which are higher than the assumed P90 scenario in the ex ante analysis.

(iii) The ex ante analysis assumed the avoided climate damage as $68.8 in 2016 prices per avoided ton of carbon dioxide (tCO_2) fixed for the duration of the project and a grid emission factor of 0.67 tCO_2 per megawatt-hour.[8] ADB adopted in 2017 a standard value of $36.3 per tCO_2 in 2016 prices with an increasing real value. This analysis used the cost of carbon emission in the RRP and adopted the new ADB standard in the sensitivity analysis.

C. Valuation of Economic Benefits

7. CONFIDENTIAL INFORMATION DELETED.

D. Valuation of Economic Costs

8. CONFIDENTIAL INFORMATION DELETED.

E. Recalculation of Economic Internal Rate of Return and Net Present Value

9. CONFIDENTIAL INFORMATION DELETED.

Table A6.1: Recalculation of Economic Internal Rate of Return

CONFIDENTIAL INFORMATION DELETED.

F. Sensitivity Analysis

10. CONFIDENTIAL INFORMATION DELETED.

Table A6.2: Sensitivity to Key Variables

CONFIDENTIAL INFORMATION DELETED.

[8] Government of Cambodia, Ministry of Environment; and Institute for Global Environmental Strategies. 2011. *Grid Emission Factor of the Phnom Penh Electricity Grid*. Phnom Penh; and Hayama, Japan.

ENVIRONMENTAL AND SOCIAL IMPACTS

A. Project Overview

1. Sunseap Asset (Cambodia) Co. Ltd. is a joint venture between Sunseap International Pte. Ltd., a Singapore-registered a subsidiary of solar developer Sunseap Group Pte. Ltd., and SchneiTec Co. Ltd, a Cambodia-registered solar power developer. Sunseap Energy (Cambodia), another joint venture between Sunseap International and SchneiTec, leased the land for the project; was the project's engineering, procurement, and construction contractor; and is the project's operation and maintenance (O&M) contractor.

2. The project involved the construction and operation of the first utility-scale solar power project in Cambodia with capacity of 10 megawatts.[1] The project site covers a total area of 207,316 square meters and is in the Chrak Mtes Commune, between the Thlok (east) and Bat Sloek (west) communes in Bavet District, Bavet City, Svay Rieng Province, near the Viet Nam border. The site is surrounded by paddy fields, and the closest residential area is about 700 meters to the east (Thlok). The closest natural protected area is the Trapeang Lpeou Wildlife Sanctuary in Takeo Province (about 122 kilometers southwest of the project site). The project site and the surrounding area are managed as part of the Bavet District.

3. The site's perimeter is secured by a chain-link fence. The site has one single-story office building, a gated main entrance (with an adjacent security guard building), a single-story switch room building, five transformers and associated shipping containers fitted out as electrical rooms, a surface water retention pond, and a covered material storage area for broken and unused solar panels in addition to other miscellaneous materials. In addition, an on-site retention pond was built in the northern portion of the site to assist in attenuating rainwater runoff for the project.

4. In compliance with the Asian Development Bank (ADB) Safeguard Policy Statement (2009), the project was classified as category B for the environment, category C for involuntary resettlement, and category C for indigenous peoples. As such, an initial environmental and social examination (IESE) was required and prepared before project approval. The IESE included an environmental and social management plan (ESMP) that Sunseap had to implement to address the project's potential environmental and social impacts and risks. ADB understands that the IESE was submitted to the Ministry of Environment for approval. The scope of work for ADB's technical advisor during due diligence also included consideration of safeguards and social matters.

5. The project's environment, health, and safety performance; and compliance with ADB Safeguard Policy Statement requirements were assessed through a desktop review of the annual monitoring reports and discussions with Sunseap. An extended annual review report mission was fielded for this project because of regional travel restrictions related to the coronavirus disease (COVID-19) pandemic response.

B. Observations

6. **Environment, health, and safety management systems.** CONFIDENTIAL INFORMATION DELETED.

[1] Asian Development Bank. 2016. *Report and Recommendation of the President to the Board of Directors: Proposed Loans and Administration of Loan to Sunseap Asset (Cambodia) Co. Ltd. for the Cambodia Solar Project in Cambodia*. Manila.

7. **Compliance with environment, health, and safety regulations, notifications, and standards.** CONFIDENTIAL INFORMATION DELETED.

8. CONFIDENTIAL INFORMATION DELETED.

9. **Environment, health, and safety monitoring.** CONFIDENTIAL INFORMATION DELETED.

10. **Land.** The project did not cause involuntary displacement impacts, and no distinct and vulnerable indigenous peoples groups were directly or indirectly affected. Sunseap identified the site when it prepared its tender documentation for Electricité du Cambodge (EDC). The project feasibility analysis included a comparison of sites that are closer to the Chrak Mtes substation but cost almost double the land purchased for the project. A key decision-making factor about the chosen site was that much of the land was not or rarely used for agriculture because of absent landowners and recurrent flooding caused by its lower elevation. All land for the project site was purchased by an agent from 27 sellers through negotiation on a willing-buyer, willing-seller basis. The land procurement process was assessed as part of the IESE and involved meeting with sellers on 24 August 2016 in Thlok. Both the IESE and lender's technical advisor (LTA) found that the purchase followed national laws and regulations.

11. An overhead transmission line for electricity evacuation used an existing canal and roadside right-of-way, which EDC selected to avoid involuntary displacement impacts. It runs about 5.5 kilometers and comprises 105 single concrete poles. CONFIDENTIAL INFORMATION DELETED.

12. **Labor.** Land sellers consulted by the IESE consultant at Thlok in 2016 said that they were not interested in working on the site, preferring to continue rice farming. Nobody at the meeting cited job opportunity as a major project benefit, but participants believed the project would bring more general development opportunities to nearby villages. They appreciated the project's proximity for potential ancillary benefits, such as nearby road and irrigation canal improvements and a steady and potentially cheaper supply of electricity. The LTA's first construction monitoring report noted that the engineering, procurement, and construction contractor had about 83 workers on-site in May 2017, 185 in June, and 250 on 13 July 2017; sex-disaggregated data were not provided. The plant became operational in October 2017, and since then the O&M contractor required only five O&M people (three engineers and two operators) and two guards during the day, and one guard to watch the site at night. In addition, four workers maintain the grounds and one cleans the solar panels regularly. Sunseap has not required the O&M contractor to prioritize hiring local women or follow its human resource management policies.

Cambodia's transition to clean energy. A video from 2019 shows one woman cleaning panels at the ADB-financed solar power plant in Bavet (video by the Asian Development Bank).

13. CONFIDENTIAL INFORMATION DELETED.

C. Conclusion

14. CONFIDENTIAL INFORMATION DELETED.